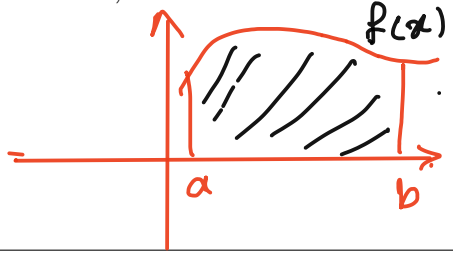
CHAPTER 6 APPLICATIONS OF INTEGRATION

6.1 Areas between Curves

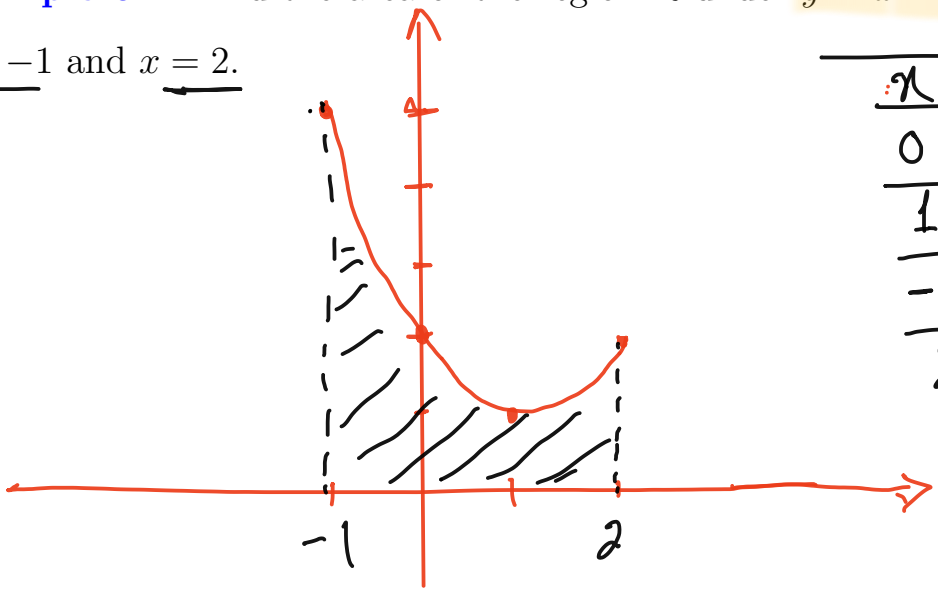
- **Region Above the x-Axis**

Let $y = f(x)$ be continuous and nonnegative on $a \le x \le b$, and let R be the region bounded by the curve $y = f(x)$, the lines $x = a$, $x = b$ and the x axis. The area of the region R is

$$A = \int_a^b f(x)dx$$



Example 6.1. Find the area of the region R under $y = x^4 - 2x^3 + 2$ between $x = -1$ and $x = 2$.



x	y
0	2
1	1
-1	5
2	2

$$\text{Area} = \int_{-1}^{2} f(x)\,dx = \int_{-1}^{2} (x^4 - 2x^3 + 2)\,dx$$

$$= \left[\frac{1}{5}x^5 - \frac{2}{4}x^4 + 2x\right]_{-1}^{2} = \cdots\cdots = \frac{51}{10}$$

Second Method :-

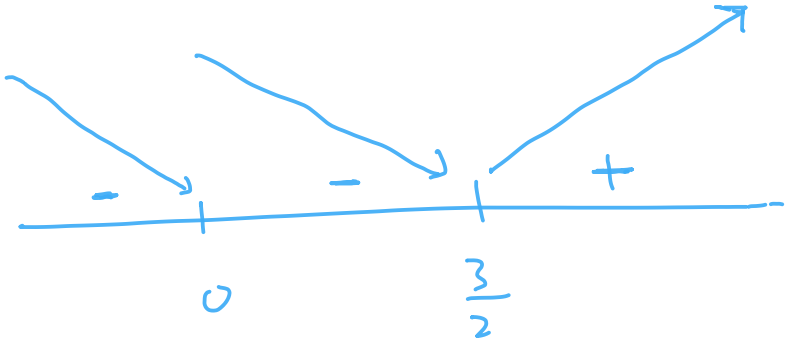
First derivative test,

$$\frac{dy}{dx} = 4x^3 - 6x^2$$

When $\frac{dy}{dx} = 0$,

$$4x^3 - 6x^2 = 0 \iff 2x^2(2x-3) = 0$$

$$\boxed{x = 0} \quad \text{or} \quad \boxed{x = \frac{3}{2}}$$



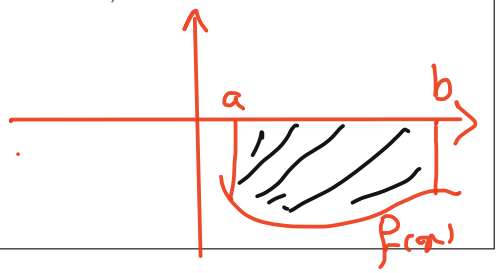
$$f\left(\frac{3}{2}\right) = \frac{5}{16} > 0.$$

$$\therefore y = x^4 - 2x^3 + 2 > 0 \quad \forall x \in \mathbb{R}$$

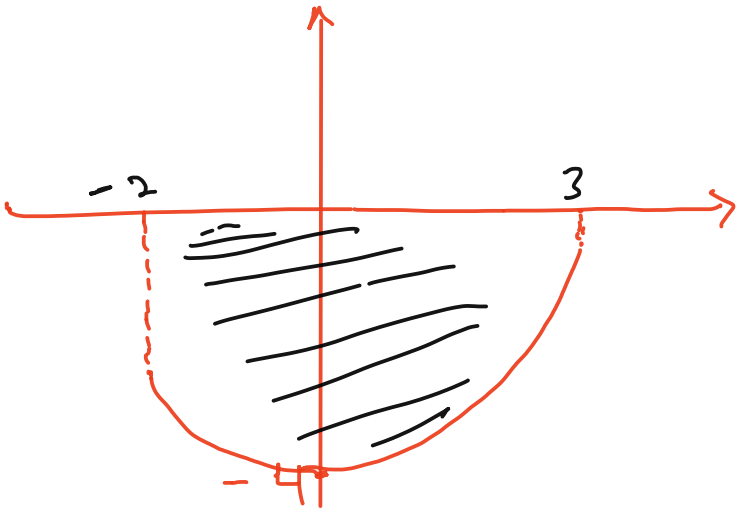
- **Region Below the x-Axis**

Let $y = f(x)$ be continuous and non-positive on $a \leq x \leq b$, and let R be the region bounded by the curve $y = f(x)$, the lines $x = a$, $x = b$ and the x axis. The area of the region R is

$$A = -\int_a^b f(x)dx$$



Example 6.2. Find the area of the region R bounded by $y = \frac{x^2}{3} - 4$, the x axis, $x = -2$ and $x = 3$.



$$\text{Area} = -\int_{-2}^{3} f(x)\,dx$$

$$= -\int_{-2}^{3} \left(\frac{x^2}{3} - 4\right) dx$$

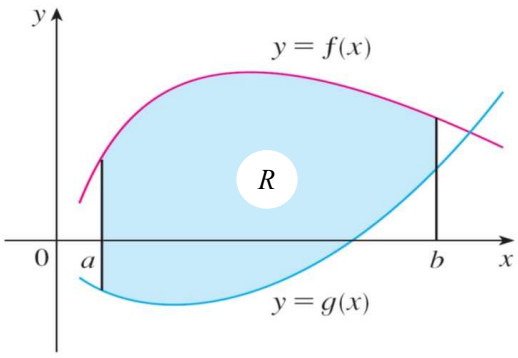
$$= -\left[\frac{1}{9}x^3 - 4x\right]_{-2}^{3}$$

$$= -\left[-\frac{145}{9}\right] = \frac{145}{9}$$

- **A Region Between Two Curves**

Consider the curves $y = f(x)$ and $y = g(x)$ with $g(x) \le f(x)$ on $a \le x \le b$. Let R be the region bounded between the curves $y = f(x)$, $y = g(x)$ and the lines $x = a$ and $x = b$. The area of R is

$$A = \int_a^b [f(x) - g(x)]\, dx = \int_a^b (y_T - y_B)\, dx$$



Example 6.3. Find the area of the region between the curves $y = x^2$ and $y = 2x - x^2$.

Intersection points:

$y = x^2$ & $y = 2x - x^2$

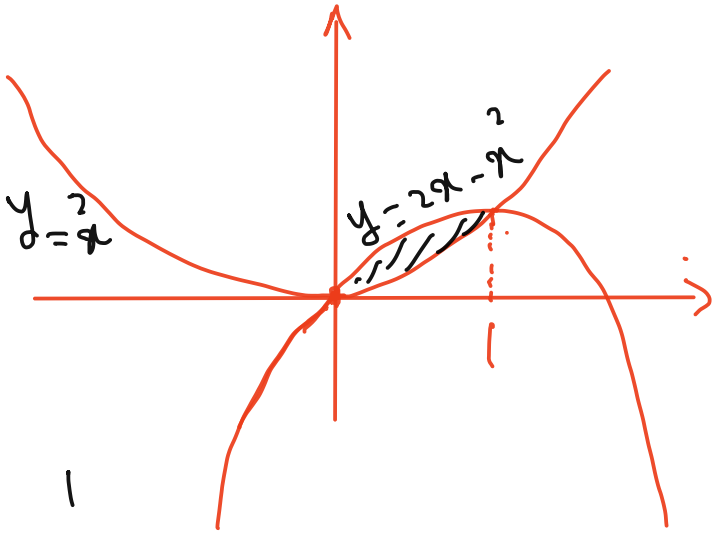
$=$

$x^2 = 2x - x^2$

$2x^2 - 2x = 0$

$2x(x-1) = 0$

$x = 0$ or $x = 1$



$$\text{Area} = \int_0^1 (2x - x^2) - (x^2)\, dx$$

$$= \int_0^1 2x - x^2 - x^2\, dx = \int_0^1 (2x - 2x^2)\, dx$$

$$= \left[x^2 - \frac{2}{3}x^3\right]_0^1 = \frac{1}{3}$$

HW

Example 6.4. Find the area of the region between the curves $y = x^4$ and $y = 2x - x^2$.

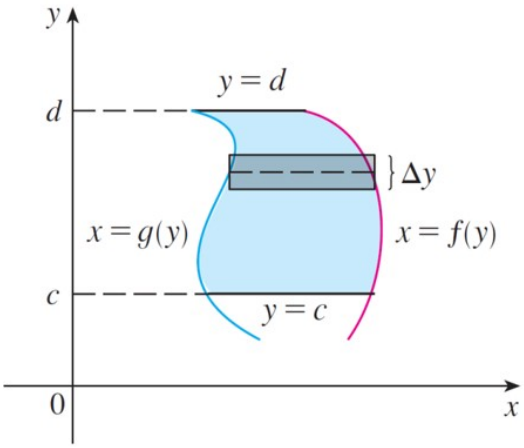
$x = 0 \; , \; x = 1$

$\text{Area} = \frac{7}{15}$

- **Horizontal Slicing**

Consider the curves $x = f(y)$ and $x = g(y)$ with $g(y) \le f(y)$ on $c \le y \le d$. Let R be the region bounded between the curves $x = f(y)$, $x = g(y)$ and the lines $y = c$ and $y = d$. The area of R is

$$A = \int_c^d [f(y) - g(y)]\, dy = \int_c^d (x_R - x_L)\, dy$$



Example 6.5. Find the area of the region bounded between the parabola $y^2 = 4x$ and the line $4x - 3y = 4$.

$y^2 = 4x \iff x = \frac{1}{4}y^2 \to (1)$

$4x - 3y = 4 \iff x = \frac{4 + 3y}{4} \to (2)$

Intersection points:

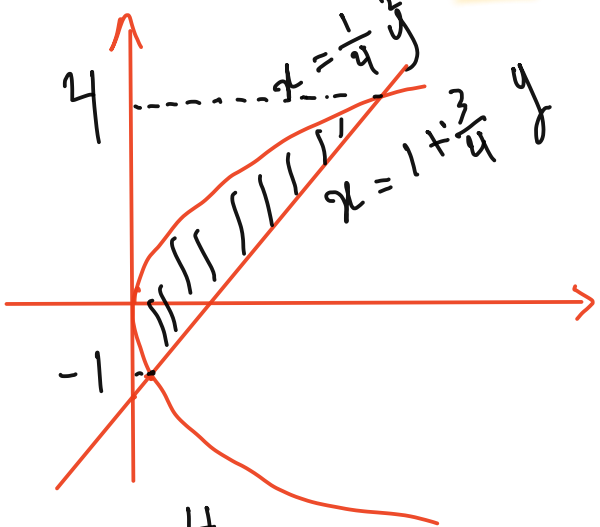
$\frac{1}{4}y^2 = \frac{4+3y}{4}$

$y^2 = 4 + 3y$

$y^2 - 3y - 4 = 0$

$(y-4)(y+1) = 0$

$y = 4$ or $y = -1$



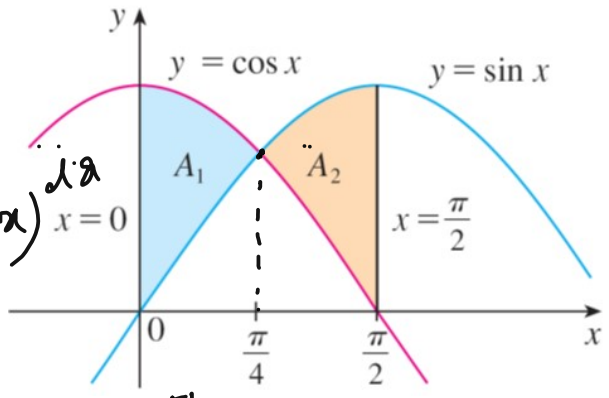
$$\text{Area} = \int_{-1}^{4} \left(1 + \frac{3}{4}y\right) - \left(\frac{1}{4}y^2\right) dy$$

$$\text{Area} = \int_{-1}^{4} \left(1 + \frac{3}{4}y - \frac{1}{4}y^2\right) dy$$

$$= \left[y + \frac{3}{8}y^2 - \frac{1}{12}y^3\right]_{-1}^{4}$$

$= \frac{125}{24}$

Example 6.6. Find the area of the region bounded by the curves $y = \sin x$, $y = \cos x$, $x = 0$ and $x = \frac{\pi}{2}$.



$$\text{Area} = A_1 + A_2$$

$$= \int_0^{\frac{\pi}{4}} (\cos x - \sin x)\,dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x - \cos x)\,dx$$

$$= \left[\sin x + \cos x\right]_0^{\frac{\pi}{4}} + \left[-\cos x - \sin x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= 2\sqrt{2} - 2$$

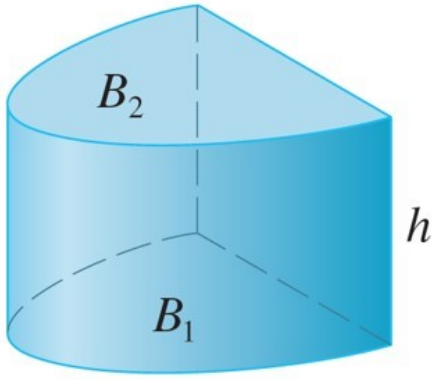
Exercises Page 434 1, 3, 7, 13, 15, 17, 21, 25, 27, 57, 61.

6.2 Volumes

- The volume of a prism is

$$V = A \cdot h$$

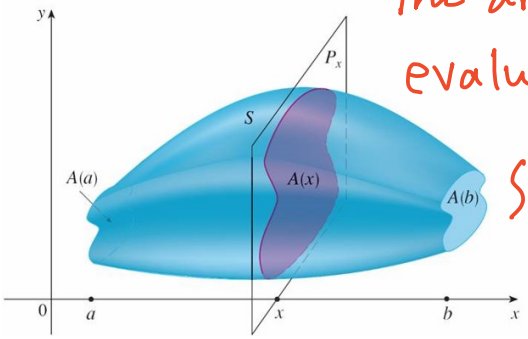
where A is the area of the base and h is the height.



- For a general solid,

$$V = \int_a^b A(x)dx$$

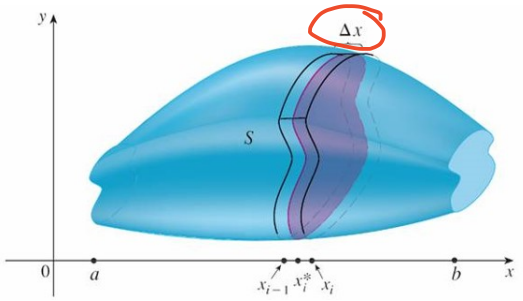
The area of cross section is a function evaluated at $x_i^* \Rightarrow A(x_i^*)$

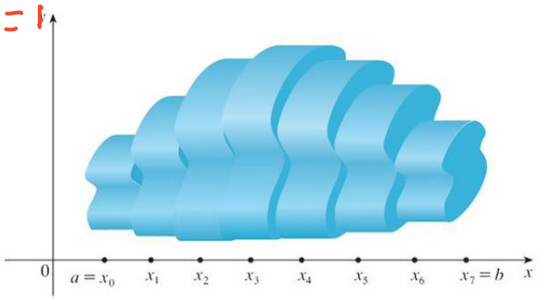


So, the volume of that slice is $A(x_i^*)\Delta x$

∴ The volume of entire solid is:

$$V \approx \sum_{i=1}^{n} A(x_i^*)\,\Delta x$$

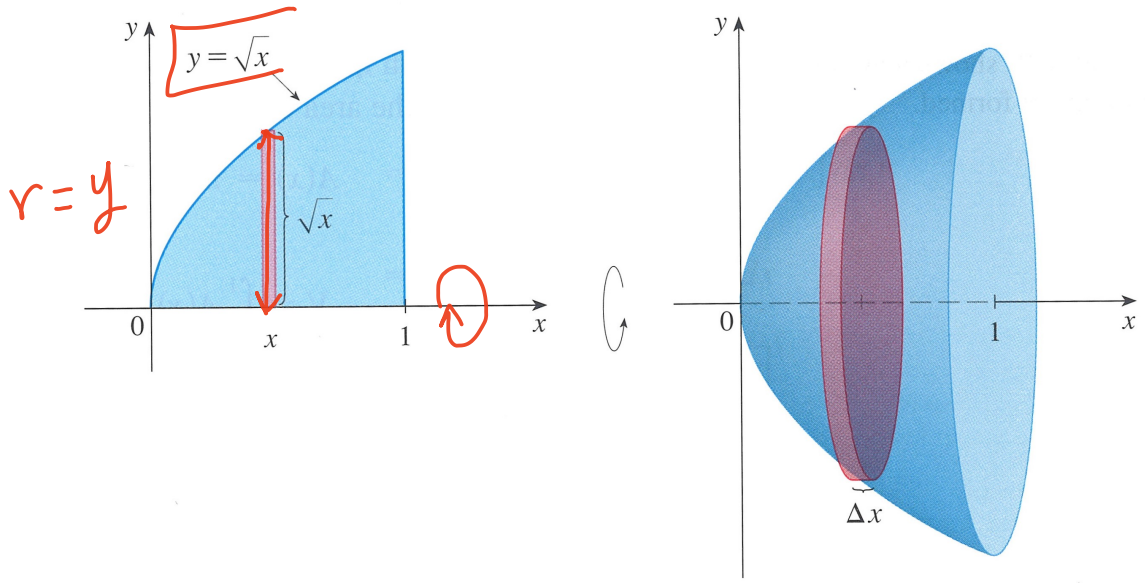


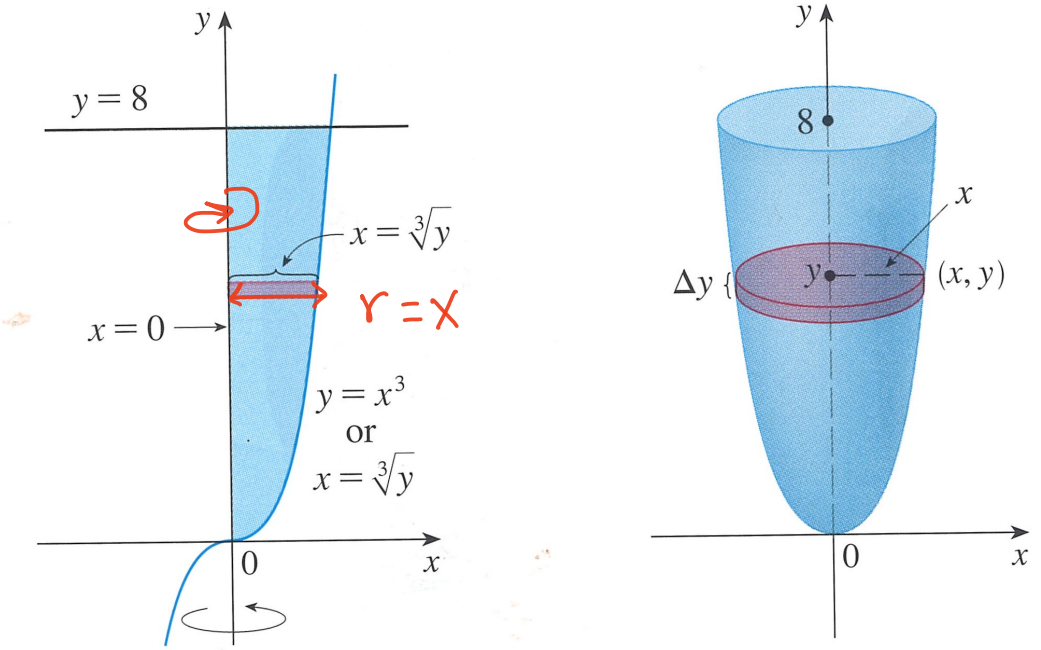


The exact volume is:

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*)\,\Delta x = \int_a^b A(x)(dx \text{ or } dy) = \int_a^b \pi r^2 (dx \text{ or } dy)$$

- When a plane region, lying entirely on one side of a fixed line in its plane, is revolved about that line, it generates a solid of revolution.





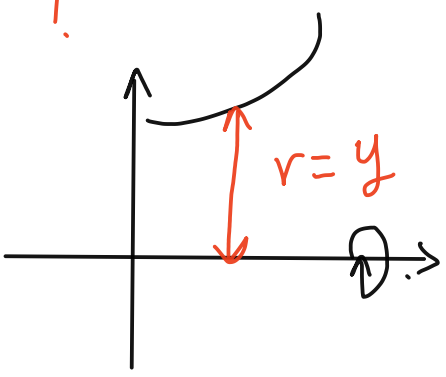
Volumes

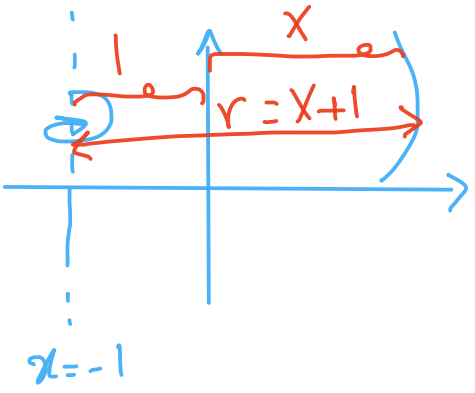
There are three methods to find the volume

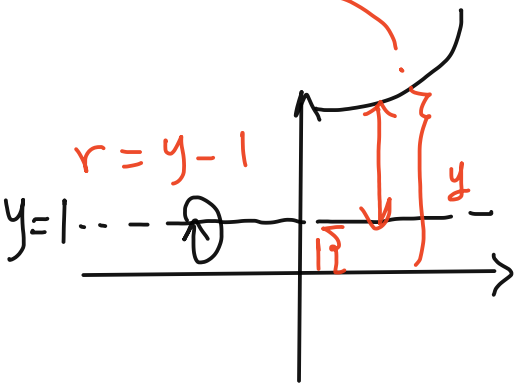
① Disk ② Washer ③ Cylindrical shell

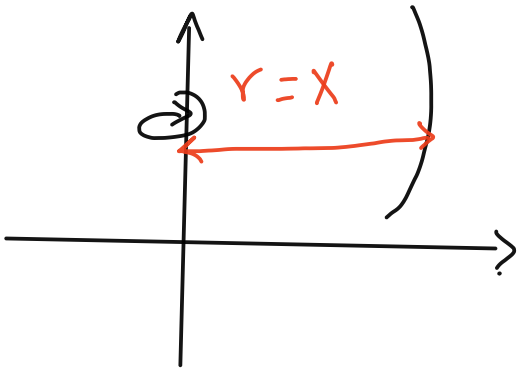
The radius (r) is the distance between the function and the axis of rotation

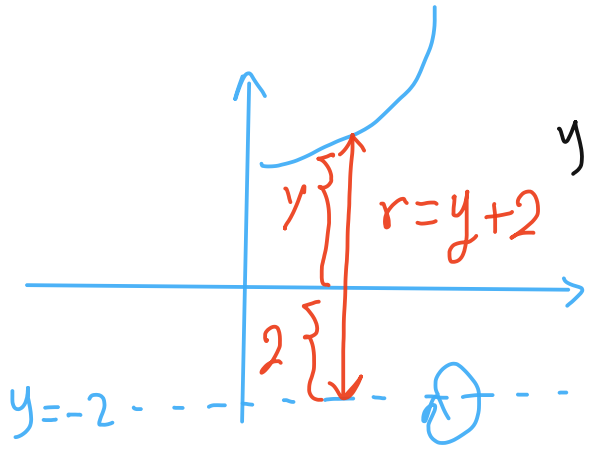
Axis of rotation

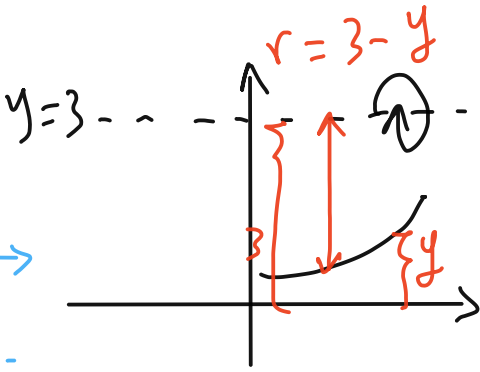


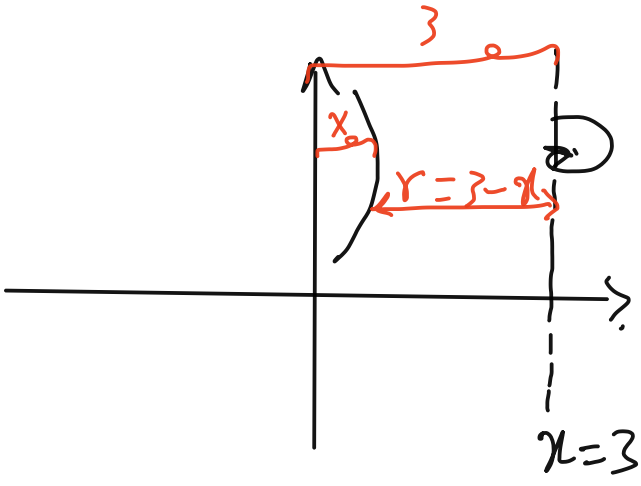


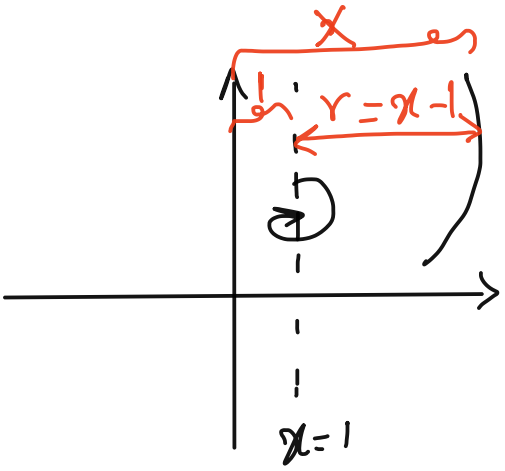




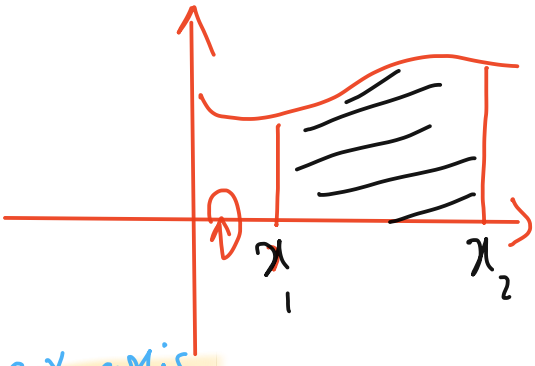








Disk Method



Case 1

If the rotating about x-axis

or any other axis parallel to the x-axis

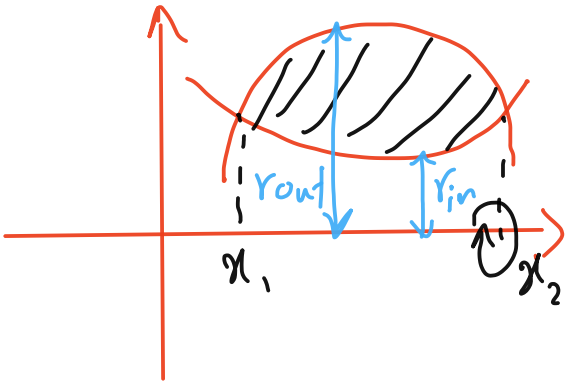
$$V = \pi \int_{x_1}^{x_2} r^2 \, dx$$

Case 2

If the rotating about y-axis or any other axis parallel to y-axis

$$V = \pi \int_{y_1}^{y_2} r^2 \, dy$$

Washer Method:



r_{out} is the furthest distance

r_{in} is the nearst distance

Case 1:

If the rotating about x-axis or any other axis

Parallel to x-axis

$$V = \pi \int_{x_1}^{x_2} \left(r_{out}^2 - r_{in}^2 \right) dx$$

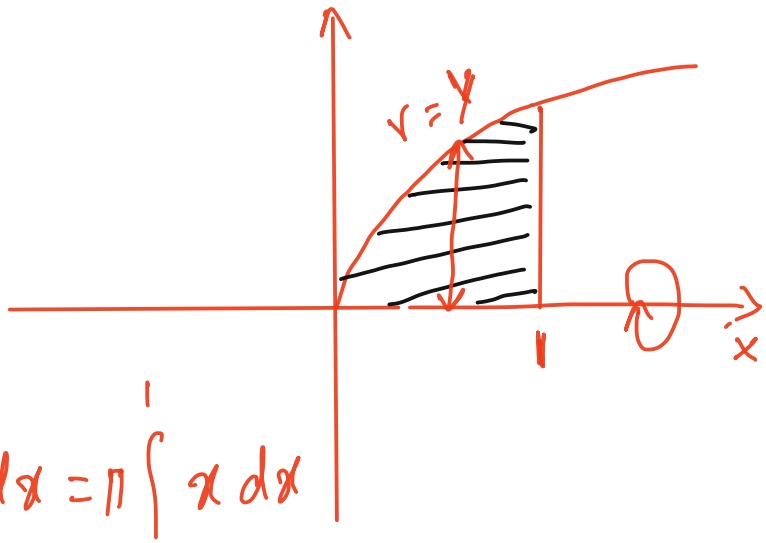
Case 2:

If the rotating about y-axis or any other axis

is parallel to y-axis

$$V = \pi \int_{y_1}^{y_2} \left(r_{out}^2 - r_{in}^2 \right) dy$$

Example 6.7. Find the volume of the solid of revolution obtained by revolving the plane region R bounded by $y = \sqrt{x}$, the x-axis, and the line $x = 1$ about the x-axis.



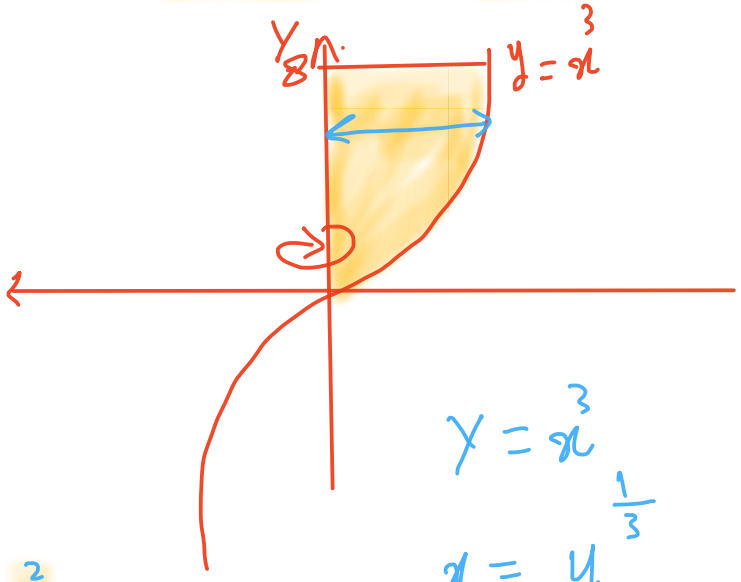
Disk Method

$$V = \pi \int_0^1 r^2 \, dx$$

$$= \pi \int_0^1 y^2 \, dx = \pi \int_0^1 (\sqrt{x})^2 \, dx = \pi \int_0^1 x \, dx$$

$$= \frac{\pi}{2} \left[x^2 \right]_0^1 = \frac{\pi}{2} (1 - 0) = \frac{\pi}{2}$$

Example 6.8. Find the volume of the solid generated by revolving the region bounded by the curve $y = x^3$, the y-axis and the line $y = 8$ about the y-axis.



Disk Method

$$V = \pi \int_0^8 r^2 \, dy$$

$$= \pi \int_0^8 x^2 \, dy$$

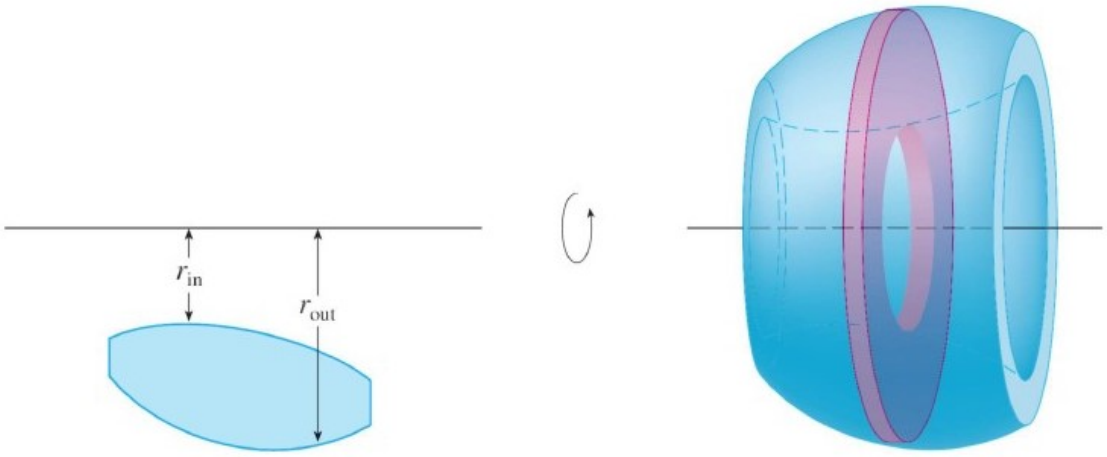
$$= \pi \int_0^8 (y^{\frac{1}{3}})^2 \, dy = \pi \int_0^8 y^{\frac{2}{3}} \, dy$$

$$= \pi \frac{3}{5} \left[y^{\frac{5}{3}} \right]_0^8 = \cdots = \frac{96}{5} \pi$$

$$y = x^3$$

$$x = y^{\frac{1}{3}}$$

- Sometimes, slicing a solid of revolution results in disks with holes in the middle. We call them washers.



Example 6.9. Find the volume of the solid generated by revolving the region bounded by the parabola $y = x^2$ and $y^2 = 8x$ about the x-axis.

Washer Method:

$$V = \pi \int_0^2 \left(r_{out}^2 - r_{in}^2 \right) dx$$

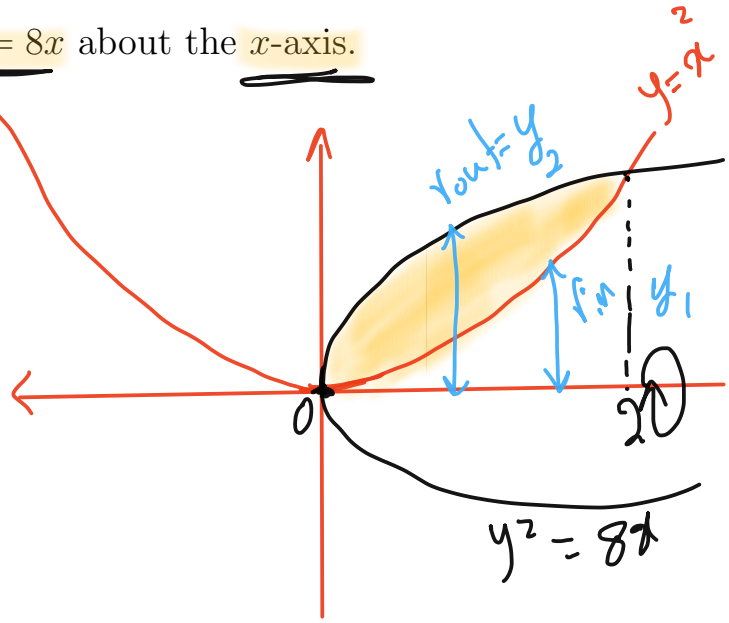
$$= \pi \int_0^2 \left(y_2^2 - y_1^2 \right) dx$$

$$= \pi \int_0^2 \left[\left(\sqrt{8x}\right)^2 - \left(x^2\right)^2 \right] dx$$

$$= \pi \int_0^2 \left(8x - x^4\right) dx$$

$$= \pi \left[\frac{8}{2} x^2 - \frac{1}{5} x^5 \right]_0^2 = \cdots$$

$$= \frac{48}{5} \pi$$



Intersection points:

$y = x^2$, $y = \pm\sqrt{8x}$

$x^2 = \sqrt{8x} \Leftrightarrow x^4 = 8x$

$x^4 - 8x = 0$

$x(x^3 - 8) = 0$

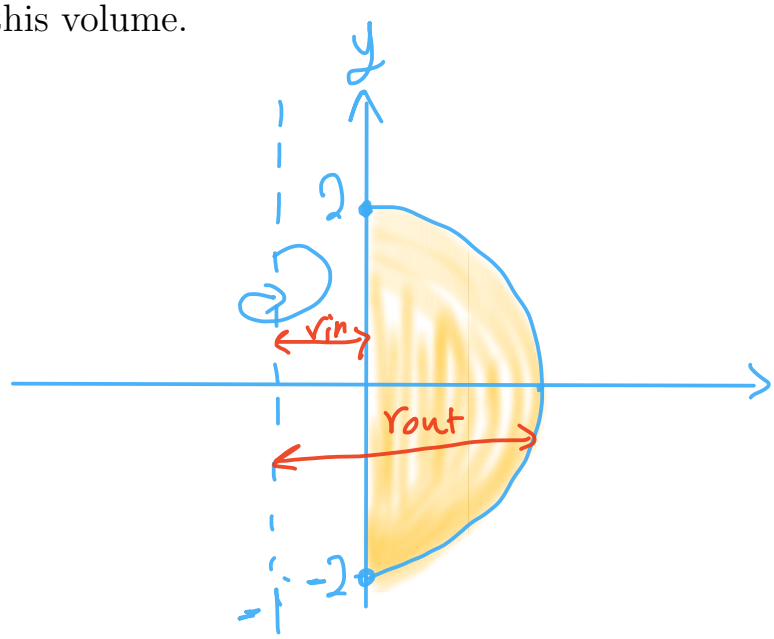
$x = 0$ or $x^3 = 8$

$x = 2$

$x^2 = 4 - y^2$

$x^2 + y^2 = 4$

Example 6.10. The semicircular region bounded by the curve $x = \sqrt{4 - y^2}$ and the y-axis is revolved about the line $x = -1$. Set up the integral that represents its volume. Then compute this volume.



Washer Method

$$V = \pi \int_{-2}^{2} \left(r_{out}^2 - r_{in}^2 \right) dy$$

$$V = \pi \int_{-2}^{2} \left[(x+1)^2 - (1)^2 \right] dy$$

$$= \pi \int_{-2}^{2} \left[\left(\sqrt{4-y^2} + 1 \right)^2 - 1 \right] dy$$

Intersection points

$x = \sqrt{4-y^2}$, $x = 0$

$\sqrt{4-y^2} = 0$

$$= \pi \int_{-2}^{2} \left[4 - y^2 + 2\sqrt{4-y^2} + 1 - 1 \right] dy$$

$$= \pi \int_{-2}^{2} \left(4 - y^2 + 2\sqrt{4-y^2} \right) dy$$

$$= \pi \left[4y - \frac{1}{3} y^3 \right]_{-2}^{2} + 2\pi \int_{-2}^{2} \sqrt{4-y^2}\, dy$$

$$= \pi \left[4y - \frac{1}{3} y^3 \right]_{-2}^{2} = 2\pi \left[\frac{\pi}{2} r^2 \right], \; r = 2$$

$$= \pi \left[\left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) \right] + 4\pi^2$$

$$= \frac{32}{3}\pi + 4\pi^2$$

Example 6.11. Let the base of a solid be the first quadrant plane region bounded by $y = 1 - \frac{x^2}{4}$, the x-axis, and the y-axis. Suppose that the cross sections perpendicular to the x-axis are squares. Find the volume of the solid.

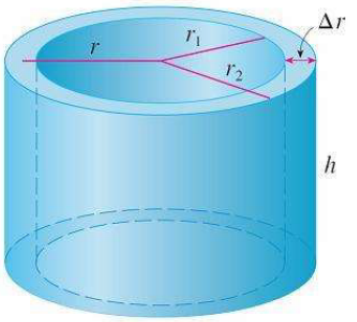
Example 6.12. The base of a solid is the region between one arch of $y = \sin x$ and the x-axis. Each cross section perpendicular to the x-axis is an equilateral triangle sitting on its base. Find the volume of the solid.

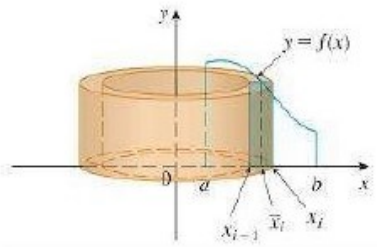
Exercises Page 446 5, 7, 9, 11, 13, 15, 17, 47, 49, 53, 55, 63.

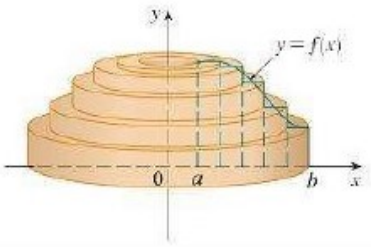
6.3 Volumes by Cylindrical Shells

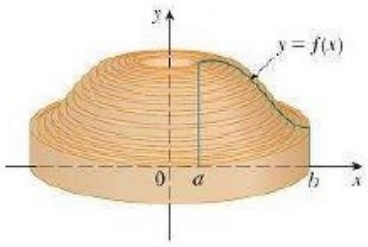
- A cylindrical shell is a solid bounded by two concentric right circular cylinders. If the inner radius is r_1 and the outer radius is r_2, and the height is h, then the volume of the cylindrical shell is

$$V = \pi \left(r_2^2 - r_1^2\right) h$$









Example 6.13. The region bounded by $y = \dfrac{1}{\sqrt{x}}$, the x-axis, $x = 1$ and $x = 4$ is revolved about the y-axis. Find the volume of the resulting solid.

Example 6.14. Find the volume of the solid generated by revolving the region in the first quadrant that is above the parabola $y = x^2$ and below the parabola $y = 2 - x^2$ about the y-axis.

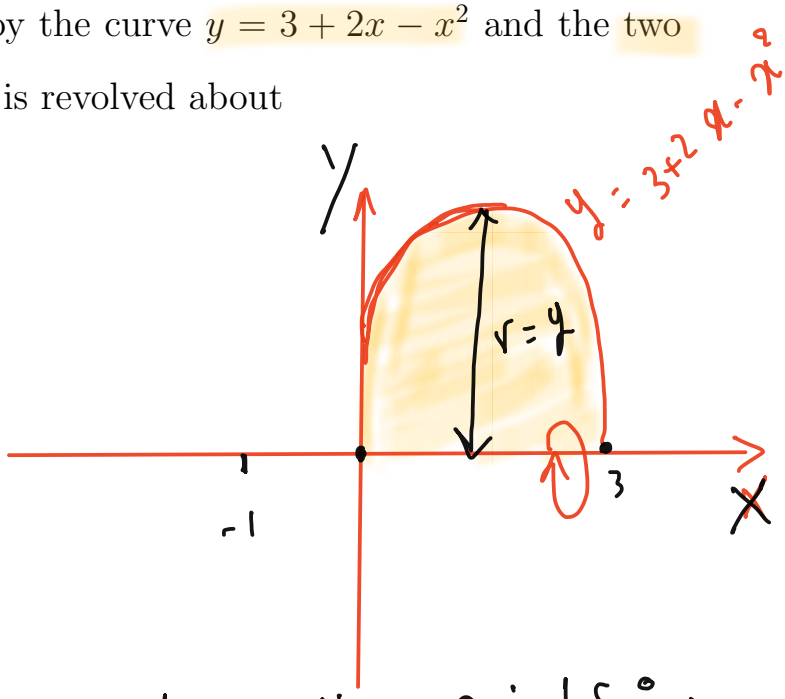
Example 6.15. Set up and evaluate an integral for the volume of the solid that results when the region R bounded by the curve $y = 3 + 2x - x^2$ and the two coordinate axes in the first quadrant is revolved about

(a) the x-axis,

(b) the y axis,

(c) the line $y = -1$,

(d) the line $x = 4$.



@ Disk Method

$$V = \pi \int_0^3 r^2 \, dx$$

$$V = \pi \int_0^3 y^2 \, dx$$

$$V = \pi \int_0^3 (3+2x-x^2)^2 \, dx$$

$$V = \pi \int_0^3 (9+12x-2x^2-4x^3+x^4) \, dx$$

$$V = \pi \left[9x + 6x^2 - \frac{2}{3}x^3 - x^4 + \frac{1}{5}x^5 \right]_0^3 = \cdots = \frac{153}{5}\pi$$

Intersection points:

$y = 3+2x-x^2$ & $y = 0$

$3+2x-x^2 = 0$

$x^2 - 2x - 3 = 0$

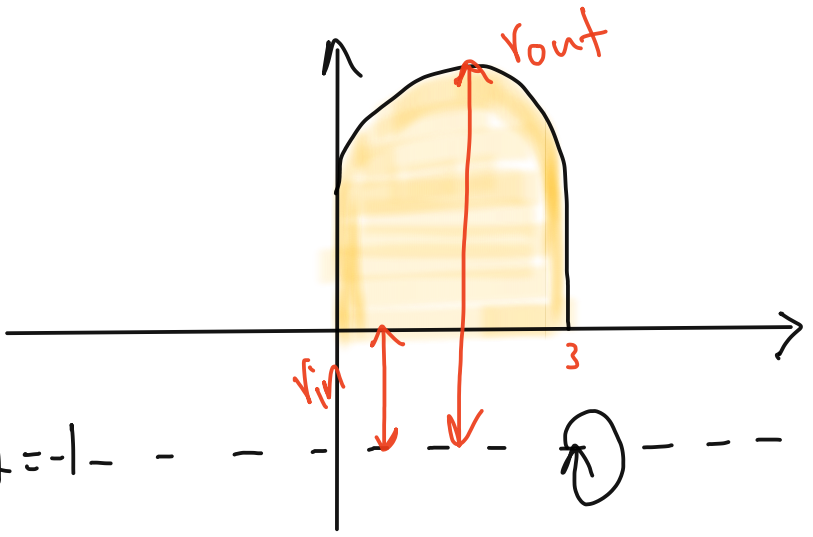
$(x-3)(x+1) = 0$

$x = 3$ or $x = -1$

Exercises Page 453 5, 7, 11, 13, 15, 17, 19, 25(a), 39, 41, 43.

Ex: 6.15

c) The line $y = -1$



Washer method

$$V = \pi \int_0^3 (r_{out}^2 - r_{in}^2)\, dx$$

$$= \pi \int_0^3 \left[(y+1)^2 - (1)^2 \right] dx = \pi \int_0^3 \left[(3+2x-x^2+1)^2 - 1 \right] dx$$

$$= \pi \int_0^3 (15 + 16x - 4x^2 - 4x^3 + x^4)\, dx$$

$$= \pi \left[15x + 8x^2 - \frac{4}{3}x^3 - x^4 + \frac{1}{5}x^5 \right]_0^3 = \cdots$$

$$= \frac{243}{5}\pi$$

6.5 Average Value of a Function

- **Average Value of Function**

If f is integrable on the interval $[a, b]$, then the average value of f on $[a, b]$ is

$$\frac{1}{b-a}\int_a^b f(x)dx$$

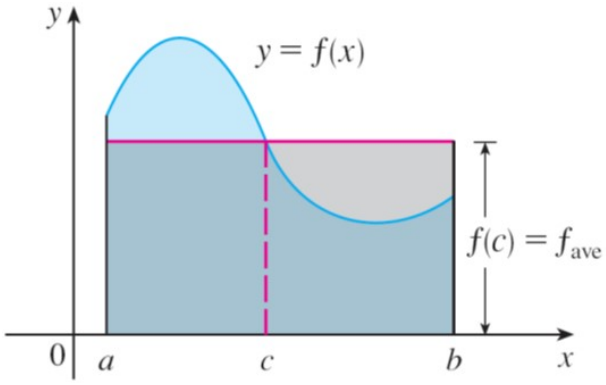
Example 6.16. Find the average value of the function $f(x) = x \sin x^2$ on the interval $[0, \sqrt{\pi}]$.

Example 6.17. Suppose the temperature in degree celsius of a metal bar of length 2 m depends on the position x according to $T(x) = 40 + 20x(2 - x)$. Find the average temperature in the bar. Is there a point where the actual temperature equals the average temperature?

- **Mean Value Theorem for Integrals**

If f is continuous on $[a, b]$, then there is a number c between a and b such that

$$f(c) = \frac{1}{b-a}\int_a^b f(x)dx$$



Example 6.18. Find all the values of c that satisfy the Mean Value Theorem for integrals for $f(x) = x^2$ on the interval $\left[-\sqrt{3}, \sqrt{3}\right]$.

Example 6.19. Find all the values of c that satisfy the Mean Value Theorem for integrals for $f(x) = \dfrac{1}{(x+1)^2}$ on the interval $[0, 2]$.

Exercises Page 463 3, 5, 7, 9(a)(b), 13.